Problem Solving through Programming (CSE103)
Multiple Choice Questions
Unit- 5

Q.No		
1`	In C, structure members are accessed using:	
	a) Dot operator (.) for variables and arrow operator (->) for pointers	b) Only arrow operator (->)
	c) Only dot operator (.)	d) None of the above
2	What will happen if you assign one structure variable to another of the same type?	
	a) All members are copied	b) Only integer members are copied
	c) Causes a runtime error	d) Not allowed in C

3	Given the structure and array declaration below, how do you access the age of the third student in the array?
	struct Student {
	char name[50];
	int age;
	};
	struct Student students[5];	

	a) students[3].age	**b) students[2].age**
	c) students.age[2]	d) students[2]->age
4	Which statement correctly defines a structure named "Person" with members "name" and "age" of type char and int respectively?	
	a) struct Person { char name; int age; }	b) struct Person { name, age; }
	c) struct { name; age; } Person;	d) struct { char* name, int age; } Person;
5	Which of the following is a collection of different data types?	
	a) String	b) Array
	c) Structure	d) Files
6	How do you declare an array of structures in C for a structure named myStruct?	
	a) array myStruct arr[];	b) myStruct arr;
	c) struct array myStruct[];	**d) myStruct array[];**
7	When a structure is passed by value to a function:	
	a) A copy of the structure is made.	b) The original structure is modified directly.
	c) Only the first member is passed.	d) This is not allowed in C.
8	Which of the following is **true** about structures in C?	
	a) All members share the same memory location.	**b) Each member has its own memory location.**
	c) The size of a structure is always the sum of sizes of its members without padding.	d) Structure members must all be of the same data type.
9	When a structure contains another structure inside it, it is called:	
	a) Self-referencing structure	**b) Nested structure**
	c) Recursive structure	d) Anonymous structure
10	Which of the following is **true** about passing structures to functions in C?	
	a) You can only pass structures by value.	b) You can only pass structures by pointer.
	c) You can pass structures both by value and by pointer.	d) Structures cannot be passed to functions.
11`	Which function is used to open a file in C?	

	a) fread()	**b) fopen()**
	c) fclose()	d) fwrite()
12	What mode opens a file for reading only?	
	a) w	**b) r**
	c) a	d) rw
13	What happens if you open a file in write (w) mode?	
	a) File is opened and contents are appended	**b) File is created if not present, existing contents are erased**
	c) File is opened for reading only	d) File is locked for editing
14	Which function is used to close a file in C?	
	a) fopen()	b) **fclose()**
	c) fread()	d) closefile()
15	What does EOF stand for?	
	a) End of file	b) Error on file
	c) End of function	d) Entry of file
16	Which function reads a line of text from a file in C?	
	a) fgets()	b) fputc()
	c) fwrite()	d) fread()
17	If you want to add data to the end of an existing file without deleting it, which mode should you use?	
	a) w	**b) a**
	c) r	d) r+
18	What happens if you try to read a file opened in write mode?	
	a) Data is read normally	b) File pointer moves to EOF immediately
	c) Error or undefined behavior	d) File is closed automatically
19	Which function reads a single character from a file in C?	
	a) fputc()	**b) fgetc()**
	c) fwrite()	d) fgets()
20	Why is it important to close a file after operations?	
	a) To speed up the program execution	**b) To free system resources and flush buffers**
	c) To open the file again later	d) To rename the file automatically